| DS | **Session** | Stack | **Question Information** | Level 1 Challenge 43 |
|---|---|---|---|---|

**Problem Description:**

You are given an array $A$ of $Q$ integers and $Q$ queries. In each query, you are given an integer $i$ $(1 \leq i \leq N)$.

Your task is to find the minimum index greater than $i$ $(1 \leq i \leq N)$ such that:

1. Sum of digits of $A_i$ is greater than the sum of digits of $A_j$
2. $A_i < A_j$

If there is no answer, then print **-1**.

**Constraints**

1 < = N, Q ≤ 10^5

1 ≤ Ai ≤ 10^9

1 ≤ Qi ≤N

**Input format**

- The first line contains two numbers $N$ and $Q$.
- The next line contains $N$ numbers.
- Next $Q$ lines contain $Q$ queries.

**Output format**

Print the answer as described in the problem

## Logical Test Cases

### Test Case 1

INPUT (STDIN)

```
5 5
62 70 28 62 92
1
5
3
4
2
```

EXPECTED OUTPUT

```
2 -1 4 -1 -1
```

### Test Case 2

INPUT (STDIN)

```
6 6
12 71 21 16 91
11
15
31
14
21
```

EXPECTED OUTPUT

```
-1 -1 -1 -1 -1 -1
```

## Mandatory Test Cases

### Test Case 1

KEYWORD

```
if(arr[x]<arr[y])
```

### Test Case 2

KEYWORD

```
if(arr2[x]>arr2[y])
```

## Complexity Test Cases

### Test Case 1

CYCLOMATIC COMPLEXITY

```
10
```

### Test Case 2

TOKEN COUNT

```
310
```

### Test Case 3

NLOC

```
57
```

## Code

solution.c

```c
#include <stdio.h>

int digitSum(int n) {
    int s = 0;
    while (n > 0) { s += (n % 10); n /= 10; }
    return s;
}

int main() {
    int n, q;
    if (scanf("%d %d", &n, &q) != 2) return 0;
    int arr[n + 1], arr2[n + 1];
    for (int i = 1; i <= n; i++) {
        scanf("%d", &arr[i]);
        arr2[i] = digitSum(arr[i]);
    }
    for (int k = 0; k < q; k++) {
        int x;
        scanf("%d", &x);
        int ans = -1;
        for (int y = x + 1; y <= n; y++) {
            if (arr[x] < arr[y] && arr2[x] < arr2[y]) {
                ans = y;
                break;
            }
        }
        printf("%d%s", ans, (k == q - 1) ? "" : " ");
    }
    printf("\n");
    return 0;
}
```

## Output

Problems Output Debug Console Terminal Ports powershell

```
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
5 5
62 70 28 62 92
1
2
3
4
5
2 -1 4 -1 -1
PS C:\C Programming>
```